FAST National University of Computer and Emerging Sciences

Karachi Campus

PROJECT PROPOSAL

Theme Park Management System

An object-oriented console application for managing visitors, rides, events, staff, feedback, and revenue

Course:	Object Oriented Programming (OOP)
Instructor:	Sir Muhammad Khalid Khan
Section:	BAI-2C
Submission Date:	April 2026
Group Members	
Member 1:	Mohammad Sarim Raza **25K-0075**
Member 2:	Abbas Ali Haider **25K-0009**

1. Project Title

Theme Park Management System — A C++ object-oriented application for managing a theme park's visitors, rides, events, staff, tickets, feedback, inventory, and revenue reports using core OOP principles.

2. Group Members

#	Name	Roll Number	Responsibility
1	Mohammad Sarim Raza	25K-0075	Class Design, Inheritance & Polymorphism
2	Abbas Ali Haider	25K-0009	File Handling, Exception Handling & Testing

3. Problem Statement

Theme parks handle many daily operations such as visitor registration, ride management, event scheduling, ticket booking, staff management, feedback collection, and revenue tracking. Manual handling of these tasks can lead to data inconsistency, poor record keeping, and difficulty in managing the park efficiently.

An object-oriented software solution is needed that models real-world park entities as classes with clear relationships, enforces business rules through encapsulation, and stores data reliably using file handling.

4. Proposed Solution

We propose to build a **Theme Park Management System** in C++ that fully utilizes OOP concepts taught in the lab. The application will run as a menu-driven console program with clearly separated classes for each entity.

Key Classes & Modules:

- **ParkEntity (Base Class):** Abstract base with ID and name — inherited by Visitor, Ride, Event, Staff, and Inventory.
- **Visitor & Ride:** Visitor stores age and email; Ride stores price and operational status.
- **Event:** Stores event name, date, capacity, and number of registered visitors.
- **Staff:** Stores staff name, role, and salary.
- **Ticket:** Records ticket booking, supports discount and output formatting.
- **Feedback:** Stores visitor feedback and rating.
- **Inventory:** Tracks items, quantity, and cost per item.
- **User:** Handles login credentials and role-based access.

- **ThemeParkSystem:** Central manager class that handles login, menus, file loading, and file saving.

5. OOP Concepts to be Implemented

OOP Concept	Where Applied
Classes & Objects	Visitor, Ride, Event, Staff, Ticket, Feedback, Inventory, User, ThemeParkSystem
Encapsulation	Private data members with getters and controlled access in all classes
Inheritance	ParkEntity → Visitor, Ride, Event, Staff, Inventory
Polymorphism	Virtual display() and serialize() overridden in derived classes
Abstract Classes	ParkEntity as the abstract base class with pure virtual methods
Operator Overloading	Ticket class overloads +, <, and <<
Templates	Generic helper functions readNumber() and findById()
Exception Handling	runtime_error and invalid_argument used for invalid operations
File Handling	All records saved and loaded using text files
STL Containers	vector, algorithm, numeric, stringstream used throughout
Friend Functions	applyDiscount() and operator<< for Ticket, calculateAverageRating() for Feedback
Static Members	Static nextId values used for automatic ID generation

6. Technology Stack

Tool / Technology	Purpose
C++ (C++17 standard)	Core programming language
Dev-C++ / Code::Blocks / VS Code	IDE for development
STL (Standard Template Library)	vector, algorithm, numeric, stringstream
File I/O (fstream)	Data persistence across sessions
Windows Console API	Colored text output in menus

7. Project Timeline

Week	Milestone	Deliverable
Week 1	Proposal & Class Design	Class structure and initial planning
Week 2	Base Classes Implementation	ParkEntity, Visitor, Ride, Event classes

Week 3	Core Logic	Ticket booking, staff, feedback, inventory
Week 4	File Handling & STL	Persistent storage and collection management
Week 5	Exception Handling & Templates	Input validation and generic helper functions
Week 6	Testing & Debugging	Functional testing and issue fixing
Week 7	Documentation & Viva Prep	Final report, code submission, viva preparation

8. Marking Scheme (Reference)

Component	Marks	Deadline
Proposal	1	Week 1
Code Submission	3	Before viva
Project Report	2	Before viva
Viva Voce	4	28 April 2026 (2C)
Total	**10**	

9. Division of Work

Member	Roll No.	Tasks Assigned
Mohammad Sarim Raza	25K-0075	Class hierarchy design, inheritance structure, polymorphism, UML diagram
Abbas Ali Haider	25K-0009	File handling, exception handling, ticket booking logic, testing and documentation

10. Expected Outcome

Upon completion, the Theme Park Management System will provide a fully working C++ console application for managing park operations with persistent data storage, clean class structure, proper use of OOP concepts, and a simple menu-driven interface for staff and manager roles.

> **Note:** All group members understand the complete project structure and can explain each class, method, and OOP concept during the viva.

Group Signature

Instructor Approval
Sir Muhammad Khalid Khan